

# System Architecture — Cornell Deep Identity Protection Platform

Layers, data flow, and the single audit commit path · v2

## CLIENT LAYER · frontend/

Static SPA served by Express — no build step, works offline in a lab

### index.html

Ingest · sign-in / register  
theme = Light | Ocean

### app.js

bootstrap · routing  
SSE listener · theme bus

### soc.js

7 SOC panels  
live risk & alerts

### user.js · portal.js

Security Center  
school portal RBAC

### forensics · bench

case timeline  
scenario benchmark

### api.js

fingerprint  
X-FP header

↓ HTTPS / JSON + SSE · X-FP fingerprint · as360\_trust cookie

## SERVER LAYER · backend/ (Express 4 · Node ≥ 24)

Single /api router · synchronous SQLite · all writes flow through commitAudit()

### routes.js

REST · SSE · static

### network.js

IP → network label

### fingerprint.js

stable device hash

### risk-engine.js

score · level · reasons[]

### config.js

weights · bands · GEO

### auth.js

auth + step-up + RBAC

### Policy decision (auth.js)

LOW ≤ 30 → allow · MEDIUM → OTP step-up · HIGH → block/approve

### commitAudit()

1 write → DB + SSE together

### totp.js · cookie.js

RFC 6238 · trust cookie

### mailer.js · utils.js

Ethereal → console

### sme.js

attack simulation A/B/C

### forensics.js

timeline · cases · export

### benchmark.js · health.js

real elapsed\_ms · counters

### portal.js

school modules · RBAC matrix

events.js — in-process pub/sub bus: **audit** · **alert** · **sim** · **cross:<uid>**

## DATA LAYER · data/authshield360.db

node:sqlite (DatabaseSync) — 23 tables · 2 versioned migrations · no native dependency

### Identity

users · devices  
trusted\_devices  
sessions  
verification\_tokens

### Risk & telemetry

audit\_trail  
success\_logins  
threat\_factors  
alerts · lists

### Forensics

cases · case\_items  
attack\_runs  
recovery\_codes  
mfa\_resets

### School portal

courses  
course\_enrollments  
assignments  
exam\_results · attendance

### Meta

schema\_migrations  
m001 school  
m002 indexes ×9  
9 audit indexes

### reset-db.js

wipe + reseed  
pristine DB  
for every  
test run

## VERIFICATION LAYER · scripts/

### smoke.js — 17

engine, no server needed  
**PASS**

### e2e-all.js — 77

live server, full journey  
**PASS**

### ui-smoke.js — 28

real browser, real DOM  
**PASS**

### restart-check.js — 18

data survives kill/restart  
**PASS**

### TOTAL — 140

140 / 140 green  
**on pristine DB**